

Introduction to Quantitative Biology

Lecture Notes: cell biology by the numbers cont. -
energy and central dogma

From Cell Biology by Numbers

September 2026

1. How much power is consumed by one human cell?

Question

What is the characteristic power consumption of one human cell?

To answer this, we need two estimates first:

1. How much power is consumed by a human?
2. How many cells are there in a human?

Once we have those two numbers, the average power consumption per cell is simply

$$P_{\text{cell}} = \frac{P_{\text{human}}}{N_{\text{cells}}}.$$

1.1 How much power is consumed by a human?

Question

What is the characteristic power consumption of a human?

- A. 10^1 W
- B. 10^2 W
- C. 10^3 W

A useful starting point is the daily food-energy intake of an adult:

$$2000 \text{ kcal/day}.$$

Using

$$1 \text{ kcal} \approx 4 \times 10^3 \text{ J},$$

the daily energy intake is

$$E_{\text{day}} \approx 2000 \frac{\text{kcal}}{\text{day}} \times 4 \times 10^3 \frac{\text{J}}{\text{kcal}} \approx 8 \times 10^6 \frac{\text{J}}{\text{day}} \sim 10^7 \frac{\text{J}}{\text{day}}.$$

The 1, few, 10 algebra

For quick order-of-magnitude estimates, it is useful to reduce numerical prefactors to three rough categories:

- numbers close to 1 are treated as 1;
- numbers of order 2 or 3 are written as **few**;
- numbers approaching 10 can be absorbed into the next power of ten.

For example,

$$1 \text{ day} = 60 \frac{\text{s}}{\text{min}} \times 60 \frac{\text{min}}{\text{h}} \times 24 \frac{\text{h}}{\text{day}}.$$

Using the 1, few, 10 algebra,

$$60 \sim 10^2, \quad 24 \sim \text{few} \times 10^1,$$

so

$$1 \text{ day} \sim 10^2 \times 10^2 \times (\text{few} \times 10^1) = \text{few} \times 10^5 \text{ s} \sim 10^5 \text{ s}.$$

Therefore,

$$P_{\text{human}} = \frac{E_{\text{day}}}{t_{\text{day}}} \sim \frac{10^7 \text{ J}}{10^5 \text{ s}} = 10^2 \text{ J s}^{-1} = 10^2 \text{ W}.$$

Hence,

$$\boxed{P_{\text{human}} \sim 100 \text{ W}}.$$

So the answer is **B.** 10^2 W .

Two useful sanity checks mentioned in class are:

- the human brain consumes roughly 20 W, about one fifth of the total;
- a rough whole-body rule of thumb is about 1 W/kg.

1.2 How many cells are there in a human?

Rather than taking the cell number as a given, we can estimate it from a characteristic cell size.

Take the linear size of a typical human cell to be roughly

$$L_{\text{cell}} \sim 10 \mu\text{m} = 10^{-5} \text{ m}.$$

Its characteristic volume is then

$$V_{\text{cell}} \sim (10 \mu\text{m})^3 = (10^{-5} \text{ m})^3 = 10^{-15} \text{ m}^3.$$

Assume that the density of a cell is approximately the density of water,

$$\rho_{\text{cell}} \sim \rho_{\text{water}} \sim 10^3 \text{ kg/m}^3.$$

Then the mass of one cell is approximately

$$m_{\text{cell}} \sim V_{\text{cell}} \rho_{\text{cell}} \sim 10^{-15} \text{ m}^3 \times 10^3 \frac{\text{kg}}{\text{m}^3} = 10^{-12} \text{ kg} = 10^{-9} \text{ g}.$$

For a 100 kg human,

$$M_{\text{human}} \sim 100 \text{ kg} = 10^5 \text{ g}.$$

Therefore,

$$N_{\text{cells}} \sim \frac{10^5 \text{ g}}{10^{-9} \text{ g/cell}} = 10^{14} \text{ cells}.$$

Thus,

$$\boxed{N_{\text{cells}} \sim 10^{14}}.$$

This is deliberately an order-of-magnitude estimate: real human cells vary greatly in size and the body is not literally composed of identical $10 \mu\text{m}$ cubes. As a sanity check, the more detailed census quoted in *Cell Biology by the Numbers* is of order 10^{13} - 10^{14} cells.

1.3 Return to the original question

Now combine the two estimates:

$$P_{\text{human}} \sim 10^2 \text{ W}, \quad N_{\text{cells}} \sim 10^{14}.$$

Therefore,

$$P_{\text{cell}} \sim \frac{10^2 \text{ W}}{10^{14} \text{ cells}} = 10^{-12} \text{ W/cell}.$$

Hence,

$$\boxed{P_{\text{cell}} \sim 10^{-12} \text{ W} = 1 \text{ pW}}.$$

So a useful order-of-magnitude picture is that an average human cell consumes power on the scale of a **picowatt**.

2. ATP as the cellular energy currency

The estimate above was made in joules and watts because we started from food energy. Inside cells, however, it is often more useful to express energetic costs in terms of ATP molecules.

2.1 How much free energy is released by one ATP hydrolysis event?

Under physiological conditions,

$$\Delta G_{\text{ATP}} \approx -50 \text{ kJ/mol} \approx -20 k_B T.$$

The negative sign indicates that ATP hydrolysis is energetically favorable and releases free energy.

Per ATP molecule,

$$\Delta G_{\text{ATP}} \sim -\frac{5 \times 10^4 \text{ J/mol}}{6 \times 10^{23} \text{ molecules/mol}} \sim -10^{-19} \text{ J/ATP}.$$

When counting how much usable energy is available per ATP, we use the magnitude

$$|\Delta G_{\text{ATP}}| \sim 10^{-19} \text{ J/ATP}.$$

2.2 How many ATP molecules does one human cell use per second?

If all of the estimated cellular power

$$P_{\text{cell}} \sim 10^{-12} \text{ J/s}$$

were converted into ATP free energy, we would obtain

$$R_{\text{ATP}} \sim \frac{10^{-12} \text{ J/s}}{10^{-19} \text{ J/ATP}} \sim 10^7 \text{ ATP/s/cell}.$$

However, this estimate started from the chemical energy contained in food. For the working estimate used in the lecture, assume that only about one third of the ingested energy is converted into usable ATP free energy:

$$\eta_{\text{ATP}} \sim \frac{1}{3}.$$

Therefore,

$$R_{\text{ATP}} \sim \eta_{\text{ATP}} \frac{P_{\text{cell}}}{|\Delta G_{\text{ATP}}|} \sim \frac{1}{3} \times 10^7 \sim \text{a few} \times 10^6 \text{ ATP/s/cell}.$$

Thus the working value for the rest of the estimate is

$$R_{\text{ATP}} \sim \text{a few} \times 10^6 \text{ ATP/s/cell}.$$

Comparison with bacteria

For rapidly growing *E. coli*, *Cell Biology by the Numbers* gives a similar order-of-magnitude ATP expenditure,

$$R_{\text{ATP}, \text{E.coli}} \sim 10^7 \text{ ATP/s},$$

corresponding to a cellular power of approximately

$$P_{\text{E.coli}} \sim 10^{-12} \text{ W/cell}$$

under the reference growth conditions considered there.

3. If a human cell spent all of its energy on protein synthesis, how long would it take to make another cell's worth of protein?

Question

If the average human cell spent all its available ATP on protein synthesis, and protein degradation were free, how long would it take to make another cell's worth of protein?

- A. a few minutes
- B. a few hours
- C. a few days
- D. a few weeks

3.1 Why use approximately 4 ATP equivalents per amino acid?

For this back-of-the-envelope calculation, we assume that the direct energetic cost of **translation/polymerization** is the dominant cost we need to count for producing protein. A useful rule of thumb is

$4 \text{ ATP equivalents/aa}$.

This estimate can be broken down as follows:

1. **Charging a tRNA with an amino acid: about 2 ATP equivalents.**

Aminoacyl-tRNA synthetase uses ATP and converts it to AMP + PP_i. Because this consumes two high-energy phosphate-bond equivalents, it is counted as approximately 2 ATP equivalents per amino acid.

2. Delivering the charged aminoacyl-tRNA to the ribosome: about 1 ATP equivalent.

This step uses one GTP during elongation (EF-Tu in bacteria; eEF1A in eukaryotes). For energetic bookkeeping, one GTP is counted as approximately one ATP equivalent.

3. Ribosome translocation by one codon: about 1 ATP equivalent.

After peptide-bond formation, the ribosome must move one codon along the mRNA. This translocation step uses one additional GTP (EF-G in bacteria; eEF2 in eukaryotes), again counted as approximately one ATP equivalent.

Therefore,

$$2 + 1 + 1 \approx 4 \text{ ATP equivalents/aa.}$$

Peptide-bond formation itself does not require an additional ATP or GTP hydrolysis in this bookkeeping; it uses the energy stored in the activated aminoacyl-tRNA bond. Initiation and termination costs are also neglected because, for a protein hundreds of amino acids long, they are small compared with the repeated elongation cost.

3.2 Estimate the time from required energy divided by the energy-consumption rate

Use the following rules of thumb:

$$4 \frac{\text{ATP}}{\text{aa}}, \quad 300 \frac{\text{aa}}{\text{protein}},$$
$$\text{a few} \times 10^6 \frac{\text{proteins}}{\mu\text{m}^3}, \quad 10^3 \frac{\mu\text{m}^3}{\text{cell}},$$

and

$$R_{\text{ATP}} \sim \text{a few} \times 10^6 \frac{\text{ATP}}{\text{cell s}}.$$

The ATP required to synthesize one cell's worth of protein is estimated as

$$\text{required energy} \sim \left(4 \frac{\text{ATP}}{\text{aa}}\right) \left(300 \frac{\text{aa}}{\text{protein}}\right) \left(\text{a few} \times 10^6 \frac{\text{proteins}}{\mu\text{m}^3}\right) \left(10^3 \frac{\mu\text{m}^3}{\text{cell}}\right).$$

That is,

$$\text{required energy} = \frac{\text{energy}}{\text{amino acid}} \times \frac{\text{amino acids}}{\text{protein}} \times \frac{\text{protein molecules}}{\text{volume}} \times \frac{\text{volume}}{\text{cell}}.$$

The minimum synthesis time is then simply

$$\text{time} \sim \frac{\text{required energy}}{\text{energy-consumption rate}}.$$

Putting everything into one expression,

$$t_{\text{protein}} \sim \frac{(4 \text{ ATP/aa}) (300 \text{ aa/protein}) (\text{a few} \times 10^6 \text{ protein}/\mu\text{m}^3) (10^3 \mu\text{m}^3/\text{cell})}{\text{a few} \times 10^6 \text{ ATP/cell/s}} \sim 10^6 \text{ s}.$$

Using

$$1 \text{ day} \sim 10^5 \text{ s},$$

we obtain

$$\boxed{t_{\text{protein}} \sim 10 \text{ days}}.$$

So the relevant scale is roughly **ten days**. Among the choices above, the closest answer is **D. a few weeks**.

This is an optimistic lower bound: it assumes that essentially all usable ATP can be devoted to protein synthesis and that protein degradation is free, while ignoring the energetic costs of all other cellular processes.